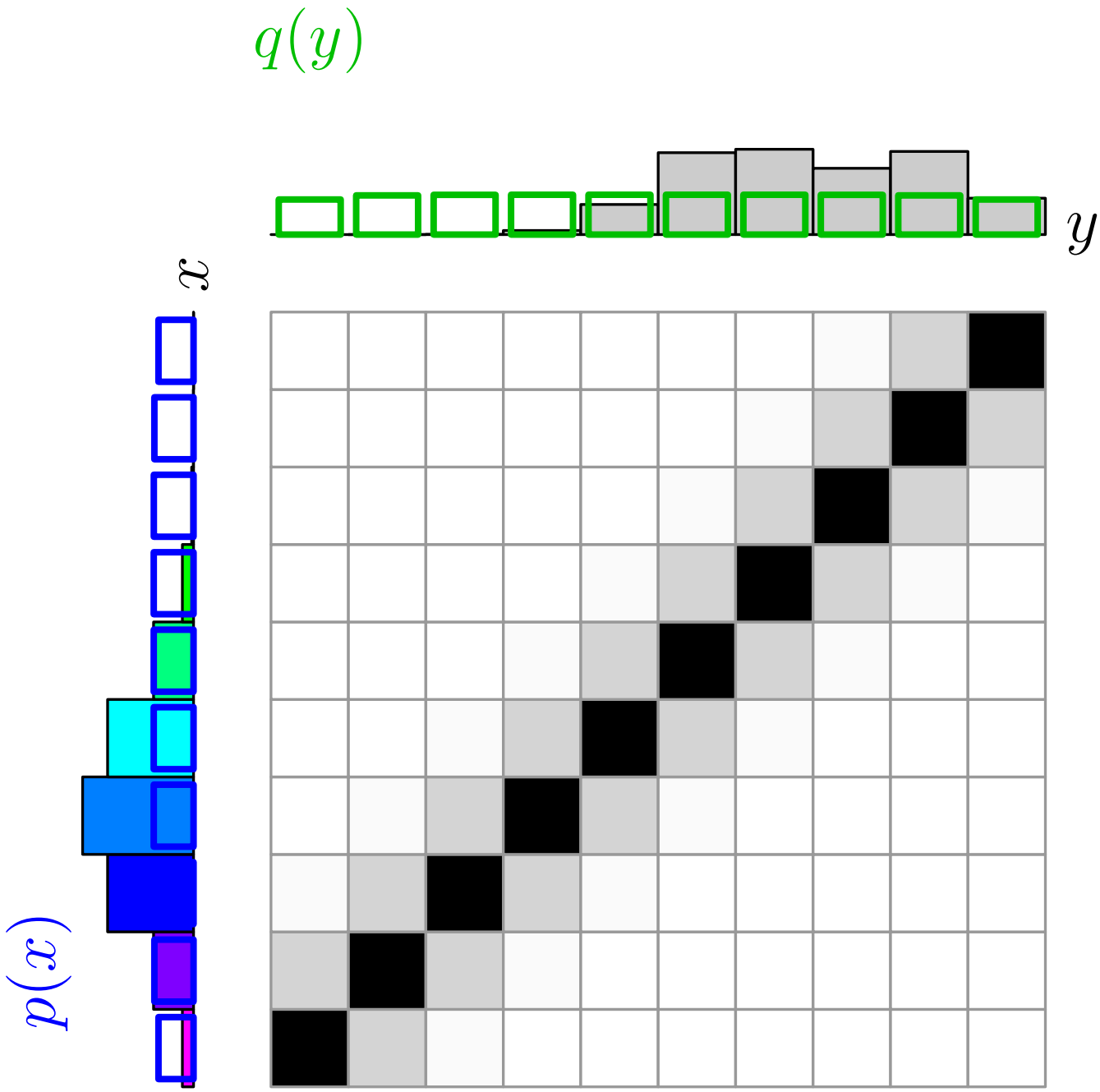

start: $\gamma \propto \mathbf{K}$

$$\boldsymbol{u} = \mathbf{1} / \sum_{ij} K_{ij},\ \boldsymbol{v} = \mathbf{1}$$

$$\gamma = \mathrm{diag}(\boldsymbol{u})\, \mathbf{K}\, \mathrm{diag}(\boldsymbol{v})$$